

Foundation (Jalapeno)

- Q1.** What is the absolute value of -5 ?
(A) -5
(B) 5
(C) 0
(D) 10
(E) 25
- Q2.** A spaceship is 12 kilometers away from the space station. What is the absolute value of its distance?
(A) -12 km
(B) 12 km
(C) 0 km
(D) 24 km
(E) 36 km
- Q3.** The temperature in a lab is -8 degrees Celsius. What is the absolute value of the temperature?
(A) -8°C
(B) 8°C
(C) 0°C
(D) 16°C
(E) 32°C
- Q4.** Evaluate the expression $|2 - 5|$.
(A) -3
(B) 3
(C) 0
(D) 7
(E) 10
- Q5.** What is the absolute value of $3 \cdot -4$?
(A) -12
(B) 12
(C) 0
(D) 16
(E) 20
- Q6.** The magnitude of a vector is given by $|3 - 2|$. What is its value?
(A) 1
(B) 5
(C) 0
(D) 11
(E) -1
- Q7.** A comet is moving at a speed of -25 km/s. What is the absolute value of its speed?
(A) -25 km/s
(B) 25 km/s
(C) 0 km/s
(D) 50 km/s
(E) 100 km/s
- Q8.** The absolute value of x is given by $|x| = 4$. What are the possible values of x ?
(A) $x = -4$
(B) $x = 4$
(C) $x = 0$
(D) $x = -4$ or $x = 4$
(E) $x = 2$ or $x = -2$

Open Ended Questions (FRQ)

- Q9.** A spacecraft is traveling from Earth to Mars. If it is 250 million kilometers away from Earth and 225 million kilometers away from Mars, what is the absolute value of the difference between its distances from the two planets? Show your work and explain your reasoning.
- Q10.** A scientist is measuring the temperature of a gas in a lab. If the initial temperature is 15°C and it decreases by 8°C , what is the absolute value of the final temperature? Explain how you would calculate this using absolute value.

Chef's Tips

- Tip 1: Ensure students understand that absolute value represents distance from zero.
- Tip 2: Emphasize the importance of considering both positive and negative values when working with absolute value.
- Tip 3: Encourage students to visualize absolute value on a number line to deepen their understanding.